

Examiner reports

Q1.

Overall students performed quite well on this question, but there were a significant minority who appeared insufficiently prepared for this topic and unable to gain much credit. In general, students performed well on parts (a) and (b), with the majority scoring full marks. Most students did not score well on part (c), failing to appreciate the significance of likely negative values within the 3 standard deviation spread. Part (d) had a very mixed response. Many students scored full marks but many others could not use the inverse normal distribution correctly.

Q4.

Many students scored full marks in part (a) by correct standardisation and use of tables. When marks were lost, it was usually in part (ii) through calculating $P(V < 5)$. There were many sound attempts at part (b), though the use of 1.645, instead of 1.96, and / or using v , instead of $5.028 + v$, lost at least 1 mark.

Q5.

The vast majority of answers here were of a very high standard, with 8 or 9 marks seen as the norm. It was also pleasing to see the absence of almost worthless attempts involving, for example, 'less than 110' implied 'less than or equal to 109 or 109.5' or the use of 2.5^2 or $\sqrt{2.5}$. When marks were lost, it was sometimes in part (b) for giving the answer to $P(V < 100)$ or, more often, in part (d) for stating 0 or 0.5 or for attempting a somewhat complicated area evaluation.

Q6.

Whilst full marks were very occasionally seen to this question, scores of 4 or less were by far the most common. Most students had little or no idea as to how to answer this question. In answering part (a), common incorrect answers were 10, $\sqrt{10}$ (3.16), 20 or 100 whereas correct answers of 5 (10 / 2) or preferably 3.33 (10 / 3) were rarely seen. Some students scored 1 mark in part (b) for giving a reasonably clear argument as to why 391 was likely to be incorrect. However, even when part (a) was answered correctly, a valid reason as to why 95.5 was likely to be incorrect proved a step too far for most students. In part (c), most students scored 1 mark for $8210 / 10 = 821$ or for $10 \times 820 = 8200$. A straight comparison of 821 with 820 scored a second mark, but those students who found 8200 were required to indicate that 8210 was within 100, not 10, for this second mark. It was very disappointing in part (c) to see those of students who failed to realise that it was

necessary to calculate $\sqrt{\frac{110}{9}}$ or $\sqrt{\frac{110}{10}}$ in order to comment on the value of $\sum (y - \bar{y})^2$, particularly given the formula for s^2 in the booklet. All too often, they argued that $\sqrt{110} \approx 10$ which, in turn, approximately equalled their (incorrect) value of σ in part (a).

Q7.

Not surprisingly, given a similar question on at least one previous paper, most students produced very sound, if not fully correct, answers in part (a) by standardising correctly in (i) and then identifying the correct related area in (ii). Some students were caught out in

(iii) by quoting zero rather than unity. In part (b), most students used Table 4 correctly to obtain $z = 1.7507$ and then solve the simple equation for x . Correct answers to part (c)(i) proved to be beyond most students. Whilst most realised that their answer to part (a)(ii) was needed for p , almost all ignored 'at least 5' and so gave an answer of p^6 .

Q8.

This question provided almost a full spread of marks. In part (a)(i), more students than in previous papers standardised 421 leading to $P(Z = 0)$ so giving a probability of 0.5 whilst others found $P(420 < X < 421)$. Answers that stated the word "impossible" only were not accepted. Answers to part (a)(ii), in which standardisation gave a positive z -value, were generally correct although a minority of students changed 425 to 424 and so lost both marks. Given that a very similar question to part (a)(iii) appeared on a recent paper, it was disappointing to see the number of incorrect answers. Most students attempted two separate standardisations rather than apply symmetry with many unable to complete the former successfully. Whilst there were fully correct answers to parts (b) & (c), there were far too many answers that scored only 1 mark for both. In the main, the loss of marks was

due to students not using Table 4 in the supplied booklet but simply equating $\frac{x - 421}{2.5}$ to $\frac{410 - \mu}{3.0}$ to 0.01 or 0.50399. Those students who, in part (c), used +2.3263, instead of -2.3263, and so obtained an answer of $\mu = 403$ scored 3 of the 4 marks.

Q9.

Most candidates scored well on this question involving normal distributions, with full marks often achieved in part (a). However, weaker candidates lost 2 marks in part (a)(ii) for calculating $P(X < 2000)$ as 0.106. Most such candidates then lost a further 2 marks in part (a)(iii) by evaluating $[(i) - (ii)]$ since, whilst this gave the correct answer of 0.864, it was the result of two errors. Answers to part (b) showed a marked improvement over similar questions on previous papers with the result that many candidates scored full marks.

Where this was not the case, it was usually through equating $\frac{1000 - 1125}{\sigma}$ to +1.2816 and then trying surreptitiously to lose the negative sign attached to σ ; a loss of 1 mark.

Q10.

Many answers to this question scored no marks and suggested that such candidates, despite completing a piece of coursework, had little experience of assessing the likely accuracy of calculations. A small minority of candidates recognised that 151.5 cm was approximately 1.5 m and so Ashley's mean value was likely to be correct. However, only

the few very best candidates realised that the standard deviation would be $\frac{2 \times 10}{4} \approx 5$ cm (a property of a normal distribution) so that 26.6 cm was likely to be incorrect. Common worthless attempts were to simply compare 26.6 cm with 10 cm or to attempt a standardisation of 151.5 using 150 and 26.6 or 10.

Q11.

Most candidates scored well on this question involving normal distributions, with full marks

often achieved in part (a). However, weaker candidates lost 2 marks in part (a)(ii) for calculating $P(X < 25)$ as 0.242. Most such candidates then lost a further 2 marks in part (a)(iii) by evaluating [(i) – (ii)] since, whilst this gave the correct answer of 0.546, it was the result of two errors. All but the weakest candidates were able to make a good, often fully correct attempt at

$$\frac{65}{1.34} - 32$$

part (b), usually by working with $\frac{65}{10}$. Some lost an accuracy mark for rounding

$\frac{65}{1.34}$ to 48 or 49, whilst others lost 2 marks for evaluating $P(\text{bill} < 65)$. A minority of candidates attempted to work with the distribution of the bill and, whilst they had the correct mean of 42.88, they left the standard deviation as 10. Answers to part (c) were poor and rarely scored the 2 marks available. It was expected, from the wording of the question, that candidates would recognise the possibility of other types of customer, fuel and / or vehicle. However, most candidates either concentrated on the actual distribution of X or commented on the variation in unleaded petrol sales due to factors already accounted for by the distribution $N(32, 10^2)$.

Q12.

Stronger students often scored full marks on this question, perhaps due to the question not requiring a qualitative response. Most students scored well in part (a) as they were usually well aware of how to standardise or how to use their calculators' in-built function. When errors did occur it was often down to rounding z -values to one decimal place thereby obtaining inaccurate answers or, particularly in part (a)(ii), finding the complement of the probability required. In part (b)(i), most students obtained 0.372 although some methods were not always clear and others were somewhat contrived. Notable among these was $[0.59871 + 0.77337 - 1]$ which could be argued to be less convincing than $[0.59871 + (1 - 0.77337)]$, particularly in a 'show that' request? Nevertheless, this and other unusual approaches that were justifiable were awarded the 2 marks available. This was not the case for those students who relied entirely on their calculators' in-built function to such an extent that they simply stated $[(P(X < 5.3) - P(X < 5.1)) \text{ or } (P(Z < 0.25) - P(Z < -0.75))] = 0.372 \dots = 0.372$. Although questions similar to part (b)(ii) have appeared fairly regularly on previous papers, many students still struggle to

distinguish between the two situations. Typical approaches were to use $\frac{0.2}{\sqrt{4}}$ in part (b)(ii)

and then use $\frac{0.5}{\sqrt{6}}$ (right!). Other less common errors in part (b)(ii) were 0.372^4 or $1 - 0.372^4$.

Q13.

Most candidates achieved the 5 marks available in this question. Almost all candidates knew how to standardise – without introducing an unnecessary continuity correction, which was penalised – and so the majority completed part (a) accurately. In part (b), the majority of candidates realised that a difference of two areas was required, with many obtaining the correct answer. Failure to apply the necessary area change for $P(Z < -0.3)$ was by far the most common error. The majority of candidates simply worked with the distribution of W , rather than $\bar{W}$, and so scored no marks. Candidates who worked with the distribution of $\bar{W}$, and so obtained correctly $P(Z > -147)$, sometimes then failed to

make the correct area change and so lost 2 marks.

Q14.

A large majority of candidates scored the 11 marks available for calculations but then struggled with the explanations. Candidates' statements in part (a) were often too vague with the key reference to 'dependence' rarely seen. Some statements confused 'arrival time' with 'time taken' whilst others introduced traffic conditions.

In part (b), almost all candidates found accurate values for b (gradient) and a (intercept) using the regression functions on their calculators and then, presented correctly the least squares regression line. A small minority of, usually able, candidates calculated regression coefficients by formulae. In almost all such cases they were successful, but may have penalised themselves regarding time available to answer other questions.

In part (c), most candidates made correct use of their equations with $x = 15$, although a significant minority used $y_{15} = 58$ from the given table of data. Apart from some candidates who mistakenly took the leaving time as 7.30 am, most proceeded, often without evidence, to a correct final answer.

As expected, it was rare to see an incorrect answer to part (d)(i). In answering part (d)(ii), candidates were much better at giving a valid statistical explanation (extrapolation) than one based on the context. Again, however, many such explanations lacked clarity, suggesting, for example, that their y -value was outside the observed range. Very few contextual reasons were sufficiently clear and detailed to be awarded a mark.

Q15.

Most candidates achieved at least 7 marks on this question, with the better candidates scoring full, or almost full, marks. Almost all candidates knew how to standardise – without introducing an unnecessary continuity correction, which was penalised – and so the majority completed part (a)(i) accurately. Again, in part (a)(ii), the majority of candidates obtained the correct answer, with failure to apply the necessary area change by far the most common error. Despite similar questions on previous papers, far too many candidates incorrectly stated 'cannot be calculated' or continued to perform a variety of lengthy worthless standardisations and subtractions of areas for the 1 mark available in part (a)(iii).

The advice to use a suitable sketch in answering part (b)(i) proved helpful, and many good attempts were presented as a result. Many candidates scored 2 of the available 3 marks for rearranging their standardisation equations but lost a mark for not explicitly stating or indicating that -1.6449 and 2.3263 were z -values. Part (b)(ii) was often answered correctly, usually by firstly eliminating μ . However, attempts that began by trying to eliminate σ were often much less successful, and those candidates who decided to solve their own, different equations gained no credit whatsoever.

Q16.

Most candidates made either a fully correct or sound attempt at part (a) and it was pleasing to see the marked absence of attempted continuity corrections or the use of

$\sqrt{2.5}$ or 2.5^2 in standardising. However, a noticeable number of candidates failed to make the necessary area change and so found $P(W < 162)$. There were many correct answers to part (b), although some valid attempts were spoilt by finding p^6 instead of p^{12} : something

that was requested on a previous paper!

Answers to part (c) were often poor with many worthless attempts, this despite similar requests on previous papers. Perhaps not surprisingly, many candidates stated 2.3263 instead of 2.5758 as the z-value. However, most candidates were unable to equate their

z-value to $\left(\frac{170 - 165}{\sigma} \right)$.

Of the minority who did, very few indeed appeared capable of solving the resultant equation for σ : a sad reflection on their algebraic skills.

Q17.

Candidates scored well on this question, with many scoring at least 9 of the 11 marks. Answers to part (a)(i) were almost invariably correct, with very few candidates stopping at

$z = 1.25$ or finding $P(Z > 1.25)$. It was gratifying to see the absence of the use of $\sqrt{0.04}$ or 0.04^2 for the standard deviation. Few candidates acted on the symmetry of the request in part (a)(ii), but instead carried out the standardising of 57.1 before subtracting two areas. However, such working was usually clear and accurate with a correct final answer. Questions similar to part (b) caused a lot of confusion on previous papers. Here confusion was much less in evidence, with a significant proportion of candidates evaluating $[(a)(i)]^{16}$ in part (b). However, perhaps on the basis of a previous paper, $[(a)(i)]^6$ was not too rare

and standardisations of the form $\frac{57.2 - 57.15}{0.04}$ or $\frac{57.2 - 57.15}{0.04/\sqrt{16}}$ were still too prevalent.

Q18.

Almost all candidates knew how to standardise, and so the overwhelming majority scored the 3 marks in part (a)(i); the remainder usually scoring 2 marks for finding $P(X > 10.5)$. In part (a)(ii), again most candidates realised that a difference in areas was required, but calculating $P(X < 10.5) - P(X > 10.0)$ was all too common, as was

the evaluation of $P\left(Z < \frac{4}{3}\right)$ as $P(Z < 1.3)$ rather than $P(Z < 1.33)$.

Answers to part (b) almost invariably involved attempts at $P(\bar{X}_6 > 10)$ instead of $(P(X > 10))^6$, this despite the fact that there was no mention of 'mean'. Consequently, the majority of candidates scored 0 marks.

Q19.

Most candidates attempting this question on the normal distribution realised that, in part (a)(i), $P(\text{not overflow}) = P(X < 155)$ and the majority then obtained the correct answer. However, a minority of candidates changed $P(X < 155)$ to $P(X \leq 154)$ — a loss of 3 marks — or performed an unnecessary area change — a loss of 1 mark. Again, in part (a)(ii), the majority of candidates stated that $P(\text{less than that printed}) = P(X < 150)$ but a significant proportion then failed to carry out the necessary area change — a loss of 2 marks.

Q20.

Most candidates achieved 3 or 5 marks in answering part (a). Almost all candidates knew how to standardise (without introducing an unnecessary continuity correction) and so the majority completed part (a)(i) accurately, but some then failed to apply the necessary area change correctly in part (a)(ii).

Candidates often scored either almost full marks or very few marks in part (b). Despite similar questions on previous papers, far too many candidates continued to perform a variety of standardisations and subtractions of areas for the 1 mark available in part (b)(i).

Few candidates answered part (b)(ii) correctly, despite a very similar request on the January 2010 paper. Some candidates with incorrect answers scored 1 mark for recognising the need to raise a probability to the power 6.

Q21.

In part (a), most candidates had at least an idea of what was required but some solutions were either partly fudged or less than entirely convincing.

In part (b), most candidates simply stated $30 - \mu = 1.6449\sigma$ rather than obtaining the equation $30 - \mu = -1.6449\sigma$; a minority attempted to correct their z -value by using $1 - 1.6449$ or $1.6449 - 1$.

Although most candidates attempted correctly, by various methods, to solve the two simultaneous equations, the aforementioned errors often resulted in impossible values for μ and σ .

Q22.

In part (a)(i), most candidates standardised the value of 5 correctly and then performed the required area change. A pleasingly large number of candidates were also successful in part (a) (ii) although $0.94520 - 0.65542 = 0.28978$ was not unusual.

In part (b), most candidates identified the correct value of z as 2.3263, but many omitted the negative sign. A small minority appeared to equate 1 per cent with 0.1. Those

candidates able to progress usually equated their z -value to $\frac{5 - (\mu \text{ or } \bar{x})}{0.05}$ but the use of 5.05 rather than 5 and/or the use of 0.025 rather than 0.05 was not unusual. In some instances the subsequent rearrangement resulted in the correct answer due to a double sign error and was penalised by mark.

Q23.

In answering part (a), most candidates realised and applied the standardisation of

$$\frac{w - 3.12}{0.08}$$

and then substituted the values of 3.20 and 2.95. Perhaps through a reluctance to draw sketches, far too many such candidates then subtracted incorrect areas. A small minority of weak candidates simply treated 3.20 and 2.95 as z -values.

Most candidates were also able to make worthwhile attempts at part (b) by equating

$$\frac{3 - 3.12}{\sigma}$$

to + 1.96. Sadly such candidates then tried to hide or ignore the fact that their

value of σ was negative. Better candidates used -1.96 or, often with explanation, changed

at an early stage to $\frac{3.12 - 3}{\sigma}$.

Q24.

Almost all candidates knew how to standardise, and there were thankfully very few instances of attempted 'continuity corrections', something that is not required in this unit. The majority completed part (a)(i) correctly, but too many did not perform the necessary area change. Whilst almost all candidates realised that a difference of areas was required in part (a)(ii), there were varying degrees of success, again often due to not making the necessary area changes — perhaps a prime example of the benefit of sketches.

In answering part (a)(iii), many candidates performed a variety of standardisations and subtractions of areas for 1 mark. Given that the request for $P(X = x \mid X \sim \text{normal})$ has appeared on previous papers, the answer should have been known. Attempts at part (b) were generally quite good with even weak candidates scoring 2 or 3 marks for equating their standardised expression to a recognisable z -value. The usual error, even by better candidates, was to use $+2.0537$ rather than -2.0537 and then to disregard or even hide the fact that the resulting standard deviation was negative.

Q25.

Almost, without exception, candidates scored the first 3 marks. In part (b), it was all too common to see the answer 0.03 instead of 0.97; a loss of 2 marks. This error was invariably compounded in part (c) where incorrect reasoning of '(a) minus (b)', instead of '{(a) minus (1 - (b))}', resulted in the correct answer but a loss of a further 2 marks.

Q26.

Almost, without exception, candidates scored the first 3 marks. In part (a)(ii), it was all too common to see candidates make an incorrect area change which resulted in the answer of 0.03 instead of 0.97; a loss of 2 marks. This error was invariably compounded in part (a)(iii) where incorrect reasoning of '(a)(i) minus (a)(ii)', instead of '{(a)(i) minus (1 - (a)(ii))}', resulted in the correct answer but a loss of a further 2 marks. Answers to part (b) were not up to the standard of those on similar questions from recent papers. All too often candidates equated a correct standardised z -statistic to $\Phi(0.025)$ or $(1 - 1.96)$ so losing 3 marks or to $+1.96$ so losing 1 mark. Incidentally, it was not unusual to see either 0.6 or 1.6, rather than 0.16, used throughout the question or, in part (a)(ii), to see 1.875 rounded to 1.86; careless errors.

Q27.

Many candidates scored around 10 marks on this question but only the highest achievers scored anywhere near the full 17 marks. In part (a)(i), almost all candidates realised the need for standardisation and so obtained $P(Z < 0.909)$. Far too many candidates then lost the final mark by quoting, from Table 3, $P(Z < 0.9)$ or $P(Z < 0.99)$. In part (a)(ii), most candidates recognised the need for an area change followed by a subtraction of areas but again lost the final mark by compounding an aforementioned error made in part (a)(i). In answering part (a)(iii), at least 50% of candidates undertook a calculation to give an answer other than zero. In part (b), many candidates scored 3 marks by showing a correct method but using $z = +1.2816$ instead of $z = -1.2816$. Perhaps a simple sketch would have shown that the value must be less than 69.5.

Answers to part (c) showed a small but welcome improvement. Whilst many candidates

considered $\frac{69.25 - 69.5}{0.55}$ and so scored 0 marks, better candidates did attempt to find the variance or standard error of $\bar{X}$ before standardising. Sadly, many such candidates then failed to find the correct area and so lost 2 marks. Again, a simple sketch would have shown that the answer was greater than 0.5.

Q28.

Many candidates scored between 8 and 12 marks on this question with only the best scoring the full 15 marks. In part (a)(i), almost all candidates realised the need for standardisation and the great majority of them completed it correctly for 3 marks. Again, in part (a)(ii), many candidates obtained the correct final answer though sometimes by unusual routes. The majority of others either introduced their answer to part (a)(i) [perhaps an idea gained from similar questions on previous papers?] or failed to perform the necessary area change and subtraction. A minority suggested that $0.8 - (1 - 0.8)$ indicated a probability of 0.6 or 0.72575. In part (b), many candidates scored 3 marks by showing a correct method but using $z = +1.0364$ instead of $z = -1.0364$. Perhaps a simple sketch would have shown that the value must be less than 140. Fewer candidates than in similar questions on previous papers used $z = 0.85$, but there was an increase in the number who simply stated 142.6, presumably from their calculators' inverse normal functions, and so scored 0 marks.

Q29.

In part (a)(i), apart from the small minority of candidates who standardised 44 (0 marks), most candidates scored all 3 marks. Again in part (a)(ii), many candidates scored all 3 marks. Those who did not score full marks usually recognised the need for 'answer (a)(i) – $P(X < 30)$ ' but then failed to make the necessary area change from $P(Z < -0.875)$ to $1 - P(Z < 0.875)$. Whilst part (b) was a routine request, too many candidates equated $\frac{45 - 40}{\sigma}$

to 0.12, 0.88 or even 0.54776, rather than 1.175 from Table 4, and so lost 3 of the 4 marks. Those candidates who used a value between 1.17 and 1.18 invariably scored 4 marks. In part (c), many candidates failed to score marks by basing their decision on means and standard deviations rather than on their calculated and given probabilities. Those who did use the latter invariably came to the correct decision even if it was stated as Route Y rather than Route B. It was disappointing to find that some candidates apparently had little, if any, knowledge of the normal distribution.

Q30.

The quality of responses to questions involving the normal distribution continued to improve with many candidates scoring at least 14 of the available 17 marks. In part (a)(i), apart from the small minority of candidates who standardised 44 (0 marks), most candidates scored all 3 marks. Again in part (a)(ii), many candidates scored all 3 marks. Those who did not, usually recognised the need for 'answer (a)(i) – $P(X < 30)$ ' but then failed to make the necessary area change from $P(Z < -0.875)$ to $1 - P(Z < 0.875)$. Whilst

part (b) was a routine request, too many candidates equated $\frac{45 - 40}{\sigma}$ to 0.12, 0.88 or even 0.54776, rather than 1.175 from Table 4, and so lost 3 of the 4 marks. In part

(c), many candidates failed to score marks by basing their decision on means and standard deviations rather than on their calculated and given probabilities. Those who did use the latter invariably came to the correct decision, even if it was stated as Route Y rather than Route B. In part (d), weaker candidates simply standardised 20 using 18 and 12 and so scored at most 1 mark. The majority of those candidates who realised that

$$SE(\bar{W}) = \frac{12}{\sqrt{36}}$$

often scored full marks, though a small minority quoted 0.841 instead of 0.159 as their final answer. In part (e), a considerable number of candidates, including some of those who had scored full marks in part (d), stated that they had not used the Central Limit Theorem as the samples had been large!

Q31.

Many candidates scored between 6 and 10 marks on this question but only the most able could accumulate more than 10 marks. In part (b), it was often the case that the only mark scored was in part (ii). Almost all candidates recognised the need to standardise with the

great majority scoring full marks. Thankfully, the use of 59 or $\sqrt{20}$ was rarer than corresponding errors on previous papers. In answering part (a)(ii), most candidates made the necessary area change before subtraction to obtain the correct answer. However, a minority of candidates evaluated 0.81594 – (answer (i)) for no reward. Given similar examples on previous papers, answers to part (a)(iii) were often disappointing. Whilst the somewhat frequent use of 1.29 (presumably from Table 3) rather than the more accurate

1.2816 from Table 4 lost only 1 mark, the all-too-frequent equating of $\frac{k - 48}{20}$ to 0.9 or $\Phi(0.9)$ lost at least 3 marks. Most candidates could not provide a valid explanation in part (b)(i). In the main answers simply stated that ‘the standard deviation was large when compared with the mean’ or even attempted to describe the practicalities of installing gas metres!

Even the few candidates who stated “negative values likely” did not justify the statement by considering, for example, the value of $\mu - 2\sigma$. Answers to part (b)(ii) usually involved “Central Limit Theorem”, “ n large” or “ $n > 30$ ” for the 1 mark available. In answering part

(b)(iii), the majority of candidates standardised 40 by $\frac{40 - 37}{25}$ and so scored no marks.

Of those that did use $\frac{40 - 37}{25 / \sqrt{35}}$ correctly, about 50% did not then perform the necessary area change.

Q32.

Statistics 1A

Most candidates scored well on this question with many scoring full marks. Somewhat surprisingly, some candidates appeared to have no requisite knowledge of the normal distribution. In part (a)(i), the standardisation was usually correct and so followed by a correct z -value and hence a correct answer from Table 3 of the supplied booklet. A few candidates standardised 59 (0 marks) or used 8 (1 mark). It was pleasing to see that a majority of candidates recognised immediately that the answer to part (a)(ii) could be obtained directly from their previous answer by subtracting 0.5. Those candidates who essentially started afresh, invariably also obtained the correct answer. Most candidates

were aware of the technique required in part (b) with many scoring all 4 marks. However, a noticeable number either equated their correct standardised value to 0.95 rather than 1.6449 or used 8 from part (a) rather than 16. In each case only 1 mark was then available.

Statistics 1B

It was disappointing to see how many candidates could not plot the four points accurately; point D causing the greatest problem. Most candidates were able to identify 'positive correlation' in part (b)(i) but slightly fewer identified the two 'anomalous' results. The vast majority of candidates identified the correct value in part (b)(ii) and identified boys C and D in part (c). However, a noticeable number did not identify the style or identified the style incorrectly whilst a small minority identified H and I presumably through thoughtlessness rather than under the impression that champion swimmers take longer times. Answers to part (d)(i) were usually correct even by those candidates not using their calculator's statistical function for r . Those candidates who identified boys other than C and D often lost most of the marks here. In answering part (d)(ii), about half the candidates merely referred to the stronger correlation and did not interpret in context.

Q33.

A pleasing number of candidates scored full marks on this last question. In part (a)(i), almost all candidates carried out a correct standardisation but a very small number then omitted the area change after referring to Table 3 in the supplied booklet. On the relatively rare occasions when marks were lost in part (a)(ii), it was often the result of subtracting 0.64058 from 0.79673. The small number of candidates who used their calculator's normal distribution function, rather than Table 3, perhaps needs to show some evidence of working so as not to automatically score zero for incorrect answers. In part (b)(i), explanations were often just sufficient to score marks. Attempts that involved mere re-arrangements of the given result scored at most 1 mark if they involved a worthwhile explanation for 1.96. Those who could do this, usually found 1.0803 from Table 4 as a start to part (b)(ii). Those candidates who were well-prepared often scored full or almost full marks. Other candidates appeared to have absolutely no idea as how to proceed beyond perhaps stating that $z = 1.0803$.

Q34.

Many candidates scored full marks on this question and it was pleasing to observe the improvements in notation and presentation. On the rare occasions when marks were lost in part (a), it was for standardising 199, 195, 199.5 or 199.9, rather than 200; something that needs to be addressed for the future. In part (b), many candidates obtained the correct final answer but not always by the most direct method; others usually failed to carry out the necessary area change. In part (c), far too many candidates assumed that all that was required was to find the difference of their two previous answers and so scored no marks. Those candidates who missed the link and so began afresh, often scored full marks. Clearly many candidates need to think more carefully about such linked parts to questions.

Q35.

For well-prepared candidates, this was a straightforward question, testing techniques that feature on the comparable papers in the legacy Specifications A and B and, as a result, they scored well.

Most candidates answered part (a)(i) correctly, with very few attempting to introduce a 'correction factor'. However, in part (a)(ii), a small minority of candidates standardised 99.5 or 99 and then these and some other candidates failed to conduct the necessary area change. Most candidates used the tables provided rather than any normal probability facility on their calculators.

In part (b), the success rate by candidates was again high. However, there was again a tendency for a small minority of candidates to use 99.5 or 99, rather than 100, in their standardisation. The only other significant error made was the use of a positive, instead of a negative, z -value.



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